Probability – 4

The probabilistic method across mathematics

Homework

11 сентября 2026 года

Theoretical minimum. The two forms of the method; the union bound, the alteration method and the local lemma; the second moment method; random vectors avoid a fixed subspace, so generic matrices have full rank; densities of integers as probabilities and equidistribution as expectation; Bernstein polynomials as expectations; Shearer's entropy inequality.

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1. (IMS 2023) Let x be a random vector of $\mathbb{R}^n$ whose coordinates are independent and equal to ± 1 with probability $\frac{1}{2}$ each, and let W be a k-dimensional subspace. Compute $\mathrm{E}\,\mathrm{dist}(x, W)^2$.

2. (Putnam 2011) Let G be an abelian group with n elements and let $\{g_1 = e, g_2, \ldots, g_k\} \subsetneq G$ be a set of distinct generators. A die selecting one of the g_i uniformly is rolled m times and the results are multiplied to give $g \in G$. Prove that there is $b \in (0,1)$ for which $\lim_{m\to\infty} b^{-2m} \sum_{x\in G} \big(\Pr(g = x) - \frac{1}{n}\big)^2$ is positive and finite.

3. (Proposed problems) Find the probability that for a positive integer n chosen at random both $\lfloor n\sqrt{2} \rfloor$ and $\lfloor n\sqrt{3} \rfloor$ are even, stating precisely in which sense the integer is chosen at random.

4. Prove Shearer's entropy inequality and deduce the Loomis–Whitney inequality: a body in $\mathbb{R}^3$ with projections of areas A_1, A_2, A_3 has volume at most $\sqrt{A_1 A_2 A_3}$.

5. Prove by the alteration method that for all k and g there is a graph of girth greater than g and chromatic number greater than k.

6. Prove the Erdős–Ko–Rado theorem by Katona's argument with a uniformly random cyclic order, and explain why the same argument fails for $k > n/2$.

7. (S.T. Yau CSMC 2019) Let $(X, \mathcal{A}, \mu)$ be a probability space and let T be measurable and measure preserving. Suppose $A \in \mathcal{A}$ satisfies $T^{-1}(A) = A$ almost everywhere. Prove that there is $B \in \mathcal{A}$ with $T^{-1}B = B$ exactly and $A = B$ almost everywhere.

8. Let A be a random $n \times n$ matrix with independent ± 1 entries. Compute $\mathrm{E}(\det A)^2$, deduce that $|\det A|$ is typically of order $\sqrt{n!}$, and compare with Hadamard's bound $n^{n/2}$.

9. Prove that for every k there is a tournament in which for any k players there is a player beating all of them, and estimate the least number of vertices for which the argument works.

10. Prove the crossing number inequality $\mathrm{cr}(G) \geqslant \frac{e^3}{64n^2}$ for $e \geqslant 4n$ by taking a random induced subgraph, and deduce the Szemerédi–Trotter bound on the number of incidences between points and lines.